

This step breaks the super increasing seq. of a_i' and will create a_i such which is not super increasing sequence.

- (5) Public Key $a = (a_1, a_2, \dots, a_n)$
 Private Key $a' = (a_1', a_2', \dots, a_n'), m, w$

• Encryption by Alice (use public key of Bob)

- (1) Use Public Key a and encrypted msg x
 $X = (x_1, x_2, \dots, x_n)$; $x_i \in \{0, 1\}$

(2)
$$S = \sum_{i=1}^n a_i x_i \leftarrow \text{C.T.}$$

 ↑
 Encryption
 o/p (sum)

• Decryption By Bob (using Private key)

- (1) Multiply S by w^{-1} w.r.t. mod m

$$\underbrace{w^{-1} S}_{\text{mod } m} = w^{-1} \sum_{i=1}^n a_i x_i \text{ mod } m$$

$$= (w^{-1} a_1 x_1 + w^{-1} a_2 x_2 + \dots + w^{-1} a_n x_n) \text{ mod } m$$

we know that,

$$a_i = (w * a_i') \text{ mod } m \quad (\because \text{from key generation})$$

$$\therefore w^{-1} a_i = a_i' \text{ mod } m \quad (\because \text{gcd}(m, w) = 1)$$

Thus,

$$w^{-1} a_i = a_i'$$

$$= (a_1'x_1 + a_2'x_2 + \dots + a_n'x_n) \bmod m$$

$$w^{-1}s \bmod m = \sum_{i=1}^n a_i'x_i \bmod m$$

Bob knows $w^{-1}s \bmod m$.

i.e. He can find it from s, w and a', m .
but x_i is not known.

Now, Bob use decryption Algo. (written previously). From that bob get $x_i \rightarrow \underline{O(n)}$.

→ Eve doesn't know a_i' . (Private key). That's why Eve requires exponential time to get P.T. (break the cryptosystem)

Ex. I/P : $a' = \{7, 15, 25, 50\}$
key generation

$$\textcircled{1} \quad m \geq \sum_{i=1}^n a_i'$$

$$\geq 7+15+25+50$$

$$\boxed{m \geq 97}$$

select $m=97$.

$\textcircled{2}$

$\textcircled{3}$ select w such that $\gcd(m, w) = 1$

let's say $w=3$

$\textcircled{4}$ compute (a_1, a_2, a_3, a_4)

$$a_1 = (w * a_1') \bmod m$$

$$= (3 * 7) \bmod 97 = 21$$

$$a_2 = (3 * 15) \bmod 97 = 45$$

$$a_3 = (3 * 25) \bmod 97 = 75$$

$$a_4 = (3 * 50) \bmod 97 = 53$$

→ Public key: $a = (21, 45, 75, 53)$

Here, $53 \neq 21 + 45 + 75 \leftarrow$ Not SIs
(Break original SIs)

→ private key: $a' = \{7, 25, 25, 50\}$

$$m = 97, w = 3$$

• Encryption by Alice

say $x = (1, 0, 1, 0) \leftarrow$ P.T. (if $A = 65$
take binary)

$$S = \sum_{i=1}^4 a_i x_i$$

$$= 21 * 1 + 45 * 0 + 75 * 1 + 53 * 0$$

$$\boxed{S = 96}$$

• Decryption by Bob

$$\begin{aligned} w^{-1} S \bmod m &= (3^{-1} * 96) \bmod 97 \\ &= [(3^{-1} \bmod 97) * 96] \bmod 97 \\ &= (65 * 96) \bmod 97 \end{aligned}$$

$$\boxed{w^{-1} S \bmod m = 32} \leftarrow \text{sum } S.$$

Now, Apply Decryption Algo

$\Rightarrow$ P/P : $\{a_1, a_2, \dots, a_n\}, S$
 O/P : $\{x_1, x_2, \dots, x_n\}$

$i = 4$ $32 \neq 50 \rightarrow \text{No}$
 $x_4 = 0$

$i = 3$ $32 \geq 25 \rightarrow \text{Yes}$
 $x_3 = 1$ $S = 32 - 25 = 7$

$i = 2$ $7 \neq 15 \rightarrow \text{No}$
 $x_2 = 0$

$i = 1$ $7 \geq 7 \rightarrow \text{Yes}$ $S = 7 - 7 = 0$
 $x_1 = 1$

$\therefore X = (x_1, x_2, x_3, x_4) = (1, 0, 1, 0)$
 which is desired result.

• Chinese Remainder Theorem (used in Rabin cryptosystem)

- used to solve a set of congruent equations with 1 variable but different modulo which are relatively prime

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

⋮

$$x \equiv a_K \pmod{m_K}$$

① Find $m = m_1 * m_2 * m_K$

② $M_1 = \frac{m}{m_1}, M_2 = \frac{m}{m_2}, M_K = \frac{m}{m_K}$

- ③ Find Multiplicative Inverse of $M_1, M_2, \dots, M_K$ using modulo $(m_1, m_2, \dots, m_K)$ call $M_1^{-1}, M_2^{-1}, \dots, M_K^{-1}$

$$\text{sol}^2 \quad x = (a_1 * M_1 * M_1^{-1} + a_2 * M_2 * M_2^{-1} + a_K * M_K * M_K^{-1}) \text{ mod } m$$

Ex

$x \equiv 2 \pmod{3}$	Verify: $23 \% 3 = 2$
$x \equiv 3 \pmod{5}$	$23 \% 5 = 3$
$x \equiv 2 \pmod{7}$	$23 \% 7 = 2$

→ Here 3, 5, 7 are relatively coprime. They have unique sol² of x .

$$a_1 = 2, a_2 = 3, a_3 = 2$$

$$m_1 = 3, m_2 = 5, m_3 = 7$$

① $m = m_1 * m_2 * m_3$
 $= 3 * 5 * 7 = \boxed{105}$

② $M_1 = \frac{m}{m_1} = \frac{105}{3} = 35$

$$M_2 = \frac{m}{m_2} = \frac{105}{5} = 21$$

$$M_3 = \frac{m}{m_3} = \frac{105}{7} = 15$$

③ $M_1^{-1} \text{ mod } m_1 = 35^{-1} \text{ mod } 3 \rightarrow \text{first perform}$
 $= 2^{-1} \text{ mod } 3$
 $= 2$
 $35 \text{ mod } 3 = 2$
 and then perform

$$M_2^{-1} \bmod m_2 = 21^{-1} \bmod 5 \\ = 1^{-1} \bmod 5 = 1$$

$$M_3^{-1} \bmod m_3 = 15^{-1} \bmod 7 = 1^{-1} \bmod 7 = 1$$

Solⁿ: $x = (a_1 * M_1 * M_1^{-1} + a_2 * M_2 * M_2^{-1} + a_3 * M_3 * M_3^{-1}) \bmod 105$

$$= (2 * 35 * 2 + 3 * 21 * 1 + 2 * 15 * 1) \\ = (140 + 63 + 30) \bmod 105 \\ = 233 \bmod 105$$

$$\boxed{x = 23}$$